TRANSVERSE BLOWOUT, LAMINAR FIRST PASSAGE, AND STABILITY INDICES IN MONOTONE SKEW PRODUCTS

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Abstract.

1. Active oscillator–actuator model

Active vibration control combines structural sensing, feedback control, and force actuation to modify the dynamic response of lightly damped mechanical structures. Direct velocity feedback is widely used because of its simplicity and effectiveness in reducing resonant vibrations [**?**, **?**, **?**]. In practical implementations, however, actuators possess finite bandwidth and internal dynamics, which may substantially modify the stability of the closed-loop system [**?**]. Moreover, actuator nonlinearities may produce self-excited oscillations and limit cycles [**?**].

Motivated by these observations, we consider the reduced-order oscillator–actuator model

$$\begin{aligned} \dot{x} &= v, \\ \dot{v} &= -\omega_0^2 x - (d-g)v + f, \\ \tau \dot{f} &= k_s x - f - \kappa x^2. \end{aligned} \tag{1}$$

Here x denotes the modal displacement, $v = \dot{x}$ the modal velocity, and f the actuator force per unit modal mass. The parameter $\omega_0 > 0$ is the natural frequency of the uncontrolled structure, $d > 0$ is the passive damping coefficient, g is the velocity-feedback gain, $\tau > 0$ is the actuator time constant, and k_s is the linear stiffness-compensation gain. The coefficient $\kappa \neq 0$ represents the leading nonlinear actuator distortion.

The actuator equation corresponds to the first-order tracking law

$$\tau \dot{f} + f = f_c(x), \qquad f_c(x) = k_s x - \kappa x^2.$$

When

$$k_s = \omega_0^2,$$

the actuator exactly compensates the structural stiffness, recovering the standard actuator model considered in active vibration control. The parameter k_s is introduced here because it provides a natural second unfolding direction for the zero–Hopf singularity.

The equilibrium points satisfy

$$v = 0, \qquad f = \omega_0^2 x,$$

and therefore

$$(k_s - \omega_0^2)x - \kappa x^2 = 0.$$

2020 *Mathematics Subject Classification.* Primary 37D25; Secondary 37A35, 37C40, 37A60.

Key words and phrases. Monotone skew products, blowout bifurcation, on–off intermittency, laminar first passage, riddled basins, stability indices, thermodynamic formalism, multifractal spectra.

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Hence

$$x = 0$$

or

$$x = \frac{k_s - \omega_0^2}{\kappa}.$$

At the critical value

$$k_s = \omega_0^2,$$

these two equilibria coalesce, and the unique equilibrium is

$$E_0 = (0, 0, 0).$$

The Jacobian matrix at E_0 is

$$A = \begin{pmatrix} 0 & 1 & 0 \\ -\omega_0^2 & g - d & 1 \\ k_s/\tau & 0 & -1/\tau \end{pmatrix},$$

whose characteristic polynomial is

$$p(\lambda) = \lambda^3 + \left(d - g + \frac{1}{\tau}\right)\lambda^2 + \left(\omega_0^2 + \frac{d-g}{\tau}\right)\lambda + \frac{\omega_0^2 - k_s}{\tau}.$$

Therefore the origin is a zero–Hopf equilibrium precisely when

$$g = d + \frac{1}{\tau}, \qquad k_s = \omega_0^2,$$

provided

$$\omega_0 \tau > 1.$$

Define

$$\Omega = \sqrt{\omega_0^2 - \frac{1}{\tau^2}}.$$

Throughout the paper we study the one-parameter unfolding

$$g = d + \frac{1}{\tau} + g_1\varepsilon, \qquad k_s = \omega_0^2 + k_1\varepsilon,$$

where g_1 and k_1 are fixed constants and $|\varepsilon| \ll 1$. This path represents a generic one-parameter slice through the codimension-two zero–Hopf bifurcation.

Theorem 1.1. *Consider system* (1). *Assume* $\omega_0 > 0, \tau > 0, \kappa \neq 0, \omega_0\tau > 1$, *and let*

$$g = d + \frac{1}{\tau} + g_1\varepsilon, \qquad k_s = \omega_0^2 + k_1\varepsilon.$$

If

$$k_1^2 - g_1^2\Omega^4\tau^2 > 0,$$

then, for every sufficiently small $\varepsilon \neq 0$, *system* (1) *possesses a periodic orbit bifurcating from the zero–Hopf equilibrium. Moreover,* $T_\varepsilon = \frac{2\pi}{\Omega} + O(\varepsilon)$ *and the periodic orbit is orbitally asymptotically stable if* $\varepsilon g_1 < 0$, *unstable if* $\varepsilon g_1 > 0$,

whereas the case $g_1 = 0$ *cannot be decided by first-order averaging.*

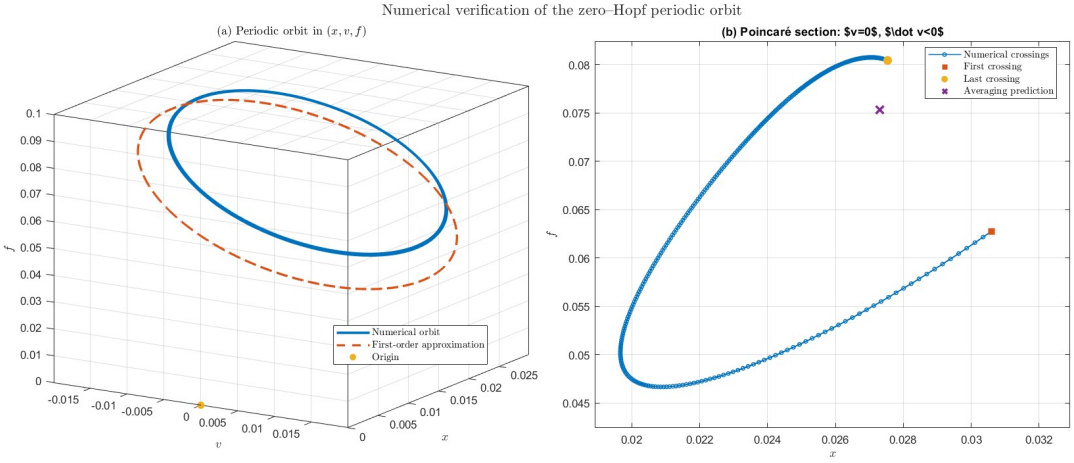


FIGURE 1. Numerical verification of the periodic orbit bifurcating from the zero–Hopf equilibrium for $\omega_0 = 2$, $\tau = 1$, $\kappa = 1$, $d = 0.2$, $g_1 = -0.2$, $k_1 = 1$, and $\varepsilon = 0.02$. Panel (a) compares the periodic orbit obtained by numerical integration of the full oscillator–actuator system with its first-order approximation derived by averaging. Panel (b) shows successive intersections of the numerical trajectory with the Poincaré section $v = 0$, $\dot v < 0$. The convergence of these intersections toward a single point provides numerical evidence for the orbital asymptotic stability predicted by $\varepsilon g_1 < 0$.

1.1. **Polar-coordinate formulation.** We now construct a real linear transformation that puts the linear part of system (1) at the critical zero–Hopf parameter values

$$g = g_* := d + \frac{1}{\tau}, \qquad k_s = k_{s*} := \omega_0^2$$

into canonical form. At the critical parameter values, the linearization of (1) at the origin is

$$A_0 = \begin{pmatrix} 0 & 1 & 0 \\ -\omega_0^2 & \dfrac{1}{\tau} & 1 \\ \dfrac{\omega_0^2}{\tau} & 0 & -\dfrac{1}{\tau} \end{pmatrix}. \tag{2}$$

Hence, the eigenvalues of A_0 are

$$\lambda_1 = 0, \qquad \lambda_{2,3} = \pm i\Omega.$$

We seek a real nonsingular matrix P such that

$$P^{-1} A_0 P = J, \qquad J = \begin{pmatrix} 0 & -\Omega & 0 \\ \Omega & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \tag{3}$$

If

$$P = \begin{pmatrix} p_1 & p_2 & p_3 \end{pmatrix},$$

then its columns must satisfy

$$A_0 p_1 = \Omega p_2, \qquad A_0 p_2 = -\Omega p_1, \qquad A_0 p_3 = 0. \tag{4}$$

Then we have

$$P = \begin{pmatrix} p_1 & p_2 & p_3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & \dfrac{1}{\omega_0^2} \\ 0 & -\Omega & 0 \\ \dfrac{1}{\tau^2} & \dfrac{\Omega}{\tau} & 1 \end{pmatrix}. \tag{5}$$

Its determinant is

$$\det P = -\frac{\Omega^3}{\omega_0^2} \neq 0. \tag{6}$$

Thus, P is nonsingular.

We introduce the real zero–Hopf coordinates (X, Y, Z) by

$$\begin{pmatrix} x \\ v \\ f \end{pmatrix} = P \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}. \tag{7}$$

Using (5), this transformation is

$$\begin{aligned} x &= X + \frac{Z}{\omega_0^2}, \\ v &= -\Omega Y, \\ f &= \frac{X}{\tau^2} + \frac{\Omega}{\tau} Y + Z. \end{aligned} \tag{8}$$

Thus the system (1) in the new variables (X, Y, Z) becomes

$$\begin{aligned} \dot{X} = & -\Omega Y + \varepsilon \left[\frac{g_1}{\Omega\tau} Y - \frac{k_1}{\Omega^2\tau} X - \frac{k_1}{\Omega^2\omega_0^2\tau} Z \right] \\ & + \frac{\kappa}{\Omega^2\tau} X^2 + \frac{2\kappa}{\Omega^2\omega_0^2\tau} XZ + \frac{\kappa}{\Omega^2\omega_0^4\tau} Z^2, \\ \dot{Y} = & \ \Omega X + \varepsilon g_1 Y, \\ \dot{Z} = & \ \varepsilon \left[-\frac{g_1\omega_0^2}{\Omega\tau} Y + \frac{k_1\omega_0^2}{\Omega^2\tau} X + \frac{k_1}{\Omega^2\tau} Z \right] \\ & - \frac{\kappa\omega_0^2}{\Omega^2\tau} X^2 - \frac{2\kappa}{\Omega^2\tau} XZ - \frac{\kappa}{\Omega^2\omega_0^2\tau} Z^2. \end{aligned} \tag{9}$$

We now transform this system into polar coordinates in order to apply the first-order averaging method.

The motion in the (X, Y)-plane is therefore a harmonic rotation with angular velocity Ω. Since we seek periodic solutions bifurcating from the zero–Hopf equilibrium, whose amplitudes are of order $O(\varepsilon)$, we introduce the scaled polar coordinates

$$X = \varepsilon r \cos\theta, \qquad Y = \varepsilon r \sin\theta, \qquad Z = \varepsilon u, \tag{10}$$

where $r \geq 0, u \in \mathbb{R}$, and $\theta \in \mathbb{S}^1$.

The scaling is chosen so that the periodic solutions obtained below remain at a distance $O(|\varepsilon|)$ from the equilibrium, which is the natural size predicted by averaging near a zero–Hopf point.

By taking θ as a new independent variable the new differential system in new variables (r, θ) as the standard first-order averaging form

$$\begin{aligned} r' &= \varepsilon F_{11}(\theta, r, u) + O(\varepsilon^2), \\ u' &= \varepsilon F_{12}(\theta, r, u) + O(\varepsilon^2), \end{aligned} \tag{11}$$

where

$$F_{11} = \frac{P\cos\theta + Q\sin\theta}{\Omega}, \qquad F_{12} = \frac{R}{\Omega}. \tag{12}$$

To simplify the notation,

$$\begin{aligned} P(r, u, \theta) &= \frac{g_1 r\sin\theta}{\Omega\tau} - \frac{k_1}{\Omega^2\tau}H + \frac{\kappa}{\Omega^2\tau}H^2, \\ Q(r, u, \theta) &= g_1 r\sin\theta, \\ R(r, u, \theta) &= -\frac{g_1\omega_0^2 r\sin\theta}{\Omega\tau} + \frac{k_1\omega_0^2}{\Omega^2\tau}H - \frac{\kappa\omega_0^2}{\Omega^2\tau}H^2. \end{aligned} \tag{13}$$

The functions F_{11} and F_{12} are 2π-periodic in the angular variable θ. Hence, system (11) is in the standard form required for the application of the first-order averaging theorem.

1.2. **First-order averaging.** The system (11) has the form

$$\begin{aligned} r' &= \varepsilon F_{11}(\theta, r, u) + O(\varepsilon^2), \\ u' &= \varepsilon F_{12}(\theta, r, u) + O(\varepsilon^2), \end{aligned} \tag{14}$$

where the functions F_{11} and F_{12} are continuously differentiable and 2π-periodic with respect to the angular variable θ.

The first averaged function associated with (14) is defined by

$$f_1(r, u) = \begin{pmatrix} f_{11}(r, u) \\ f_{12}(r, u) \end{pmatrix}, \tag{15}$$

where

$$\boxed{f_1(r, u) = \int_0^{2\pi} \begin{pmatrix} F_{11}(\theta, r, u) \\ F_{12}(\theta, r, u) \end{pmatrix} d\theta.} \tag{16}$$

The existence of periodic solutions is a direct consequence of the first-order averaging theorem.

Theorem 1.2 (First-order averaging)**.** *Consider the differential system*

$$x' = \varepsilon F(\theta, x) + O(\varepsilon^2),$$

where $F(\theta, x)$ is continuously differentiable and 2π-periodic in θ.
Suppose that the averaged function

$$f(x) = \int_0^{2\pi} F(\theta, x)\, d\theta$$

has a zero x_0 satisfying

$$f(x_0) = 0, \qquad \det Df(x_0) \neq 0.$$

Then, for all sufficiently small $|\varepsilon| > 0$, there exists a unique 2π-periodic solution

$$x(\theta, \varepsilon) = x_0 + O(\varepsilon),$$

of the averaged system. Moreover, this periodic solution corresponds to a periodic solution of the original differential system.

Therefore, our task reduces to computing the averaged function $f_1(r,u)$, determining its zeros, and verifying that the Jacobian matrix is nonsingular at the corresponding solution.

To evaluate the integrals, we use the identities

$$\int_0^{2\pi} \sin\theta\, d\theta = \int_0^{2\pi} \cos\theta\, d\theta = \int_0^{2\pi} \sin\theta\cos\theta\, d\theta = 0,$$

$$\int_0^{2\pi} \cos^3\theta\, d\theta = 0,$$

and

$$\int_0^{2\pi} \sin^2\theta\, d\theta = \int_0^{2\pi} \cos^2\theta\, d\theta = \pi.$$

Using the expressions for P, Q, and R in (13), the functions appearing in (11) are

$$\boxed{\begin{aligned} F_{11} = {} & \frac{g_1 r\sin\theta\cos\theta}{\Omega^2\tau} + \frac{g_1 r\sin^2\theta}{\Omega} \\ & - \frac{k_1\cos\theta}{\Omega^3\tau}\left(r\cos\theta + \frac{u}{\omega_0^2}\right) \\ & + \frac{\kappa\cos\theta}{\Omega^3\tau}\left(r\cos\theta + \frac{u}{\omega_0^2}\right)^2, \end{aligned}} \tag{17}$$

and

$$\boxed{\begin{aligned} F_{12} = {} & -\frac{g_1\omega_0^2 r\sin\theta}{\Omega^2\tau} \\ & + \frac{k_1\omega_0^2}{\Omega^3\tau}\left(r\cos\theta + \frac{u}{\omega_0^2}\right) \\ & - \frac{\kappa\omega_0^2}{\Omega^3\tau}\left(r\cos\theta + \frac{u}{\omega_0^2}\right)^2. \end{aligned}} \tag{18}$$

Expanding the quadratic terms gives

$$\boxed{\begin{aligned} F_{11} = {} & \frac{g_1 r\sin\theta\cos\theta}{\Omega^2\tau} + \frac{g_1 r\sin^2\theta}{\Omega} - \frac{k_1 r\cos^2\theta}{\Omega^3\tau} \\ & - \frac{k_1 u\cos\theta}{\Omega^3\tau\omega_0^2} + \frac{\kappa r^2\cos^3\theta}{\Omega^3\tau} \\ & + \frac{2\kappa r u\cos^2\theta}{\Omega^3\tau\omega_0^2} + \frac{\kappa u^2\cos\theta}{\Omega^3\tau\omega_0^4}, \end{aligned}} \tag{19}$$

and

$$\boxed{\begin{aligned} F_{12} = {} & -\frac{g_1\omega_0^2 r\sin\theta}{\Omega^2\tau} + \frac{k_1\omega_0^2 r\cos\theta}{\Omega^3\tau} + \frac{k_1 u}{\Omega^3\tau} \\ & - \frac{\kappa\omega_0^2 r^2\cos^2\theta}{\Omega^3\tau} - \frac{2\kappa r u\cos\theta}{\Omega^3\tau} - \frac{\kappa u^2}{\Omega^3\tau\omega_0^2}. \end{aligned}} \tag{20}$$

Integrating (19) over one period yields

$$f_{11}(r,u) = \frac{\pi r}{\Omega}\left(g_1 - \frac{k_1}{\Omega^2\tau} + \frac{2\kappa u}{\Omega^2\tau\omega_0^2}\right). \tag{21}$$

Similarly,

$$f_{12}(r,u) = \frac{\pi}{\Omega^3\tau}\left(2k_1 u - \kappa\omega_0^2 r^2 - \frac{2\kappa u^2}{\omega_0^2}\right). \tag{22}$$

Hence, the first averaged function is

$$\boxed{f_1(r,u) = \begin{pmatrix} \dfrac{\pi r}{\Omega}\left(g_1 - \dfrac{k_1}{\Omega^2\tau} + \dfrac{2\kappa u}{\Omega^2\tau\omega_0^2}\right) \\ \dfrac{\pi}{\Omega^3\tau}\left(2k_1 u - \kappa\omega_0^2 r^2 - \dfrac{2\kappa u^2}{\omega_0^2}\right) \end{pmatrix}}. \tag{23}$$

Unlike several classical zero–Hopf problems for which the first averaged function vanishes identically and higher-order averaging is required, the present oscillator–actuator model possesses a nontrivial first averaged function. Consequently, first-order averaging is sufficient to detect periodic solutions bifurcating from the zero–Hopf equilibrium.

1.3. **Zeros of the first averaged function.** According to Theorem 1.2, the existence of periodic solutions is determined by the zeros of the first averaged function (23). We therefore seek all solutions of

$$f_1(r,u) = 0, \qquad r > 0,$$

where the restriction $r > 0$ excludes the equilibrium itself and corresponds to genuine periodic oscillations.

Recall that

$$\begin{aligned} f_{11}(r,u) &= \frac{\pi r}{\Omega}\left(g_1 - \frac{k_1}{\Omega^2\tau} + \frac{2\kappa u}{\Omega^2\tau\omega_0^2}\right), \\ f_{12}(r,u) &= \frac{\pi}{\Omega^3\tau}\left(2k_1 u - \kappa\omega_0^2 r^2 - \frac{2\kappa u^2}{\omega_0^2}\right). \end{aligned} \tag{24}$$

Since $r > 0$, the first equation

$$f_{11}(r,u) = 0$$

is equivalent to

$$g_1 - \frac{k_1}{\Omega^2\tau} + \frac{2\kappa u}{\Omega^2\tau\omega_0^2} = 0.$$

Hence,

$$\boxed{u_0 = \frac{\omega_0^2}{2\kappa}\left(k_1 - g_1\Omega^2\tau\right).} \tag{25}$$

Substituting (25) into the second equation

$$f_{12}(r,u) = 0,$$

gives

$$2k_1u_0 - \kappa\omega_0^2 r^2 - \frac{2\kappa u_0^2}{\omega_0^2} = 0.$$

After simplification,

$$\boxed{r_0^2 = \frac{k_1^2 - g_1^2\Omega^4\tau^2}{2\kappa^2}.} \tag{26}$$

Consequently,

$$\boxed{r_0 = \sqrt{\frac{k_1^2 - g_1^2\Omega^4\tau^2}{2\kappa^2}}.} \tag{27}$$

Therefore, a nontrivial zero exists if and only if

$$\boxed{k_1^2 - g_1^2\Omega^4\tau^2 > 0.} \tag{28}$$

Under condition (28), the averaged system possesses the isolated zero

$$\boxed{(r_0, u_0) = \left(\sqrt{\frac{k_1^2 - g_1^2\Omega^4\tau^2}{2\kappa^2}},\ \frac{\omega_0^2}{2\kappa}\left(k_1 - g_1\Omega^2\tau\right)\right).} \tag{29}$$

Since $r_0 > 0$, this solution corresponds to a nontrivial oscillatory state rather than to the equilibrium point.

1.4. **Nondegeneracy of the averaged zero.** To apply Theorem 1.2, it remains to verify that the zero (29) is nondegenerate.

The Jacobian matrix of the averaged function is

$$Df_1(r,u) = \begin{pmatrix} \dfrac{\partial f_{11}}{\partial r} & \dfrac{\partial f_{11}}{\partial u} \\ \dfrac{\partial f_{12}}{\partial r} & \dfrac{\partial f_{12}}{\partial u} \end{pmatrix}.$$

Differentiating (24) yields

$$\begin{aligned} \frac{\partial f_{11}}{\partial r} &= \frac{\pi}{\Omega}\left(g_1 - \frac{k_1}{\Omega^2\tau} + \frac{2\kappa u}{\Omega^2\tau\omega_0^2}\right), \\ \frac{\partial f_{11}}{\partial u} &= \frac{2\pi\kappa r}{\Omega^3\tau\omega_0^2}, \\ \frac{\partial f_{12}}{\partial r} &= -\frac{2\pi\kappa\omega_0^2 r}{\Omega^3\tau}, \\ \frac{\partial f_{12}}{\partial u} &= \frac{\pi}{\Omega^3\tau}\left(2k_1 - \frac{4\kappa u}{\omega_0^2}\right). \end{aligned} \tag{30}$$

Evaluating these derivatives at the zero (r_0, u_0) and using (25) gives

$$\left.\frac{\partial f_{11}}{\partial r}\right|_{(r_0,u_0)} = 0,$$

and

$$\left.\frac{\partial f_{12}}{\partial u}\right|_{(r_0,u_0)} = \frac{2\pi g_1}{\Omega}.$$

Hence,

$$\boxed{Df_1(r_0,u_0)=\begin{pmatrix} 0 & \dfrac{2\pi\kappa r_0}{\Omega^3\tau\omega_0^2} \\ -\dfrac{2\pi\kappa\omega_0^2 r_0}{\Omega^3\tau} & \dfrac{2\pi g_1}{\Omega}\end{pmatrix}.} \tag{31}$$

Its determinant is

$$\begin{aligned}\det Df_1(r_0,u_0) &= \frac{4\pi^2\kappa^2 r_0^2}{\Omega^6\tau^2} \\ &= \frac{2\pi^2\left(k_1^2-g_1^2\Omega^4\tau^2\right)}{\Omega^6\tau^2}.\end{aligned} \tag{32}$$

Therefore,

$$\boxed{\det Df_1(r_0,u_0)=\frac{2\pi^2\left(k_1^2-g_1^2\Omega^4\tau^2\right)}{\Omega^6\tau^2}>0.} \tag{33}$$

Condition (28) therefore guarantees that the zero (29) is isolated and nondegenerate. Consequently, all hypotheses of Theorem 1.2 are satisfied.

1.5. **Existence and stability of the periodic orbit.** Since all the hypotheses of Theorem 1.2 are satisfied, the averaged system (11) possesses a 2π-periodic solution satisfying

$$(r(\theta,\varepsilon),u(\theta,\varepsilon))=(r_0,u_0)+O(\varepsilon),$$

where (r_0,u_0) is given by (29).

The averaging theorem implies that this periodic solution corresponds to a periodic solution of the original differential system (**??**). Consequently, for all sufficiently small $|\varepsilon|>0$, system (**??**) possesses a periodic orbit whose amplitude is $O(|\varepsilon|)$.

Poincaré-map interpretation. The existence result can be interpreted in terms of the Poincaré map.

Let

$$\Pi_\varepsilon:\mathbb{R}^2\to\mathbb{R}^2$$

denote the Poincaré map associated with one complete revolution $\theta\mapsto\theta+2\pi$ of the reduced system (11). Standard averaging theory gives the expansion

$$\boxed{\Pi_\varepsilon(z)=z+\varepsilon f_1(z)+O(\varepsilon^2),} \tag{34}$$

where

$$z=(r,u),$$

and f_1 is the first averaged function (23).

Since

$$f_1(r_0,u_0)=0,$$

equation (34) implies that

$$\Pi_\varepsilon(r_0,u_0)=(r_0,u_0)+O(\varepsilon^2).$$

Because

$$\det Df_1(r_0,u_0)\neq 0,$$

the Implicit Function Theorem guarantees the existence of a unique fixed point

$$(r_\varepsilon, u_\varepsilon) = (r_0, u_0) + O(\varepsilon)$$

of the Poincaré map. This fixed point corresponds precisely to the periodic orbit obtained from the averaging theorem.

Stability. The derivative of the Poincaré map satisfies

$$\boxed{D\Pi_\varepsilon(r_\varepsilon, u_\varepsilon) = I + \varepsilon Df_1(r_0, u_0) + O(\varepsilon^2),} \tag{35}$$

where

$$Df_1(r_0, u_0) = \begin{pmatrix} 0 & \dfrac{2\pi\kappa r_0}{\Omega^3\tau\omega_0^2} \\ -\dfrac{2\pi\kappa\omega_0^2 r_0}{\Omega^3\tau} & \dfrac{2\pi g_1}{\Omega} \end{pmatrix}.$$

Its trace and determinant are

$$\boxed{\operatorname{tr} Df_1(r_0, u_0) = \frac{2\pi g_1}{\Omega}, \qquad \det Df_1(r_0, u_0) = \frac{2\pi^2\left(k_1^2 - g_1^2\Omega^4\tau^2\right)}{\Omega^6\tau^2}.} \tag{36}$$

The eigenvalues of $Df_1(r_0, u_0)$ are

$$\boxed{\lambda_\pm = \frac{\pi g_1}{\Omega} \pm \frac{\pi}{\Omega^3\tau}\sqrt{3g_1^2\Omega^4\tau^2 - 2k_1^2}.} \tag{37}$$

Consequently, the two nontrivial Floquet multipliers of the periodic orbit satisfy

$$\boxed{\mu_i = 1 + \varepsilon\lambda_i + O(\varepsilon^2), \qquad i = 1, 2.} \tag{38}$$

(The third Floquet multiplier is identically equal to one because it corresponds to the direction tangent to the periodic orbit.)

If

$$3g_1^2\Omega^4\tau^2 < 2k_1^2,$$

the eigenvalues form a complex conjugate pair with

$$\operatorname{Re}\lambda_\pm = \frac{\pi g_1}{\Omega}.$$

Otherwise, the eigenvalues are real. Since

$$\det Df_1(r_0, u_0) > 0,$$

they necessarily have the same sign, which is determined by the trace.

Therefore,

$$\operatorname{Re}\lambda_i < 0 \quad \Longleftrightarrow \quad g_1 < 0,$$

and

$$\operatorname{Re}\lambda_i > 0 \quad \Longleftrightarrow \quad g_1 > 0.$$

Since

$$\mu_i = 1 + \varepsilon\lambda_i + O(\varepsilon^2),$$

the periodic orbit is asymptotically stable whenever

$$\varepsilon g_1 < 0,$$

and unstable whenever

$$\varepsilon g_1 > 0.$$

Hence we obtain the following result.

The averaged system possesses the simple zero (r_0, u_0), where

$$r_0 = \sqrt{\frac{k_1^2 - g_1^2\Omega^4\tau^2}{2\kappa^2}}, \qquad u_0 = \frac{\omega_0^2}{2\kappa}(k_1 - g_1\Omega^2\tau),$$

and

$$\det Df_1(r_0, u_0) \neq 0.$$

Therefore, all the hypotheses of the first-order averaging theorem are satisfied. Consequently, Theorem **??** follows.

1.6. **Approximation of the bifurcating periodic orbit.** The periodic solution obtained in Theorem **??** is expressed in the variables (r, θ, u). To recover the periodic orbit in the original coordinates, we first return to the time variable.

From (**??**),

$$\dot{\theta} = \Omega + O(\varepsilon),$$

and therefore

$$\boxed{\theta(t, \varepsilon) = \Omega t + \theta_0 + O(\varepsilon),} \tag{39}$$

where θ_0 is an arbitrary initial phase.

Moreover,

$$r(t, \varepsilon) = r_0 + O(\varepsilon), \qquad u(t, \varepsilon) = u_0 + O(\varepsilon),$$

where

$$r_0 = \sqrt{\frac{k_1^2 - g_1^2\Omega^4\tau^2}{2\kappa^2}}, \qquad u_0 = \frac{\omega_0^2}{2\kappa}\left(k_1 - g_1\Omega^2\tau\right).$$

Consequently, the leading-order period is

$$\boxed{T_\varepsilon = \frac{2\pi}{\Omega} + O(\varepsilon).} \tag{40}$$

Reconstruction in the (X, Y, Z) coordinates. Using the polar transformation

$$X = \varepsilon r\cos\theta, \qquad Y = \varepsilon r\sin\theta, \qquad Z = \varepsilon u,$$

together with (39), we obtain

$$\boxed{\begin{aligned} X(t, \varepsilon) &= \varepsilon r_0\cos(\Omega t + \theta_0) + O(\varepsilon^2), \\ Y(t, \varepsilon) &= \varepsilon r_0\sin(\Omega t + \theta_0) + O(\varepsilon^2), \\ Z(t, \varepsilon) &= \varepsilon u_0 + O(\varepsilon^2). \end{aligned}} \tag{41}$$

Hence, to leading order, the periodic orbit is a circle of radius εr_0 in the (X, Y)-plane, translated by εu_0 in the Z-direction.

Reconstruction in the original variables. Finally, recall that the inverse linear transformation is

$$x = X + \frac{Z}{\omega_0^2}, \qquad v = -\Omega Y, \qquad f = \frac{X}{\tau^2} + \frac{\Omega}{\tau} Y + Z.$$

Substituting (41) yields

$$\boxed{\begin{aligned} x(t,\varepsilon) &= \varepsilon \left(r_0 \cos(\Omega t + \theta_0) + \frac{u_0}{\omega_0^2} \right) + O(\varepsilon^2), \\ v(t,\varepsilon) &= -\varepsilon \Omega r_0 \sin(\Omega t + \theta_0) + O(\varepsilon^2), \\ f(t,\varepsilon) &= \varepsilon \left(\frac{r_0}{\tau^2} \cos(\Omega t + \theta_0) + \frac{\Omega r_0}{\tau} \sin(\Omega t + \theta_0) + u_0 \right) + O(\varepsilon^2). \end{aligned}} \tag{42}$$

Substituting the explicit expressions for r_0 and u_0, the leading-order approximation of the bifurcating periodic orbit becomes

$$\boxed{\begin{aligned} x(t,\varepsilon) &= \varepsilon \sqrt{\frac{k_1^2 - g_1^2 \Omega^4 \tau^2}{2\kappa^2}} \cos(\Omega t + \theta_0) \\ &\quad + \frac{\varepsilon}{2\kappa} \left(k_1 - g_1 \Omega^2 \tau \right) + O(\varepsilon^2), \\ v(t,\varepsilon) &= -\varepsilon \Omega \sqrt{\frac{k_1^2 - g_1^2 \Omega^4 \tau^2}{2\kappa^2}} \sin(\Omega t + \theta_0) + O(\varepsilon^2), \\ f(t,\varepsilon) &= \frac{\varepsilon}{\tau^2} \sqrt{\frac{k_1^2 - g_1^2 \Omega^4 \tau^2}{2\kappa^2}} \cos(\Omega t + \theta_0) \\ &\quad + \frac{\varepsilon \Omega}{\tau} \sqrt{\frac{k_1^2 - g_1^2 \Omega^4 \tau^2}{2\kappa^2}} \sin(\Omega t + \theta_0) \\ &\quad + \frac{\varepsilon \omega_0^2}{2\kappa} \left(k_1 - g_1 \Omega^2 \tau \right) + O(\varepsilon^2). \end{aligned}} \tag{43}$$

Therefore, whenever

$$k_1^2 - g_1^2 \Omega^4 \tau^2 > 0,$$

a family of periodic solutions bifurcates from the zero–Hopf equilibrium. The amplitude of the oscillation is proportional to $|\varepsilon|$, the leading angular frequency is Ω, and the leading period is $2\pi/\Omega$. The stability of the periodic orbit is determined by the sign of εg_1, in accordance with Theorem **??**.

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